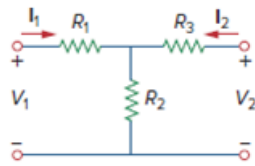


Problem

Obtain the g parameters for the wye circuit of Fig. 19.96.

Figure 19.96.



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

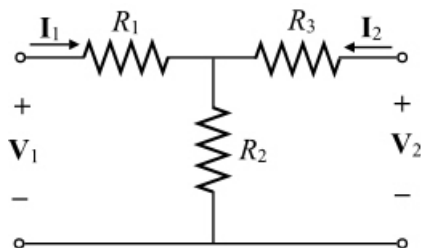


Figure 1

Step 2 of 6

Write the inverse hybrid parameters or g -parameters:

$$g_{11} = \left. \frac{I_1}{V_1} \right|_{I_2=0} \dots\dots (1)$$

$$g_{12} = \left. \frac{I_1}{I_2} \right|_{V_1=0} \dots\dots (2)$$

$$g_{21} = \left. \frac{V_2}{V_1} \right|_{I_2=0} \dots\dots (3)$$

$$g_{22} = \left. \frac{V_2}{I_2} \right|_{V_1=0} \dots\dots (4)$$

Step 3 of 6

To calculate parameters g_{11} and g_{21} , connect a test voltage source to the input port and open circuit the output port.

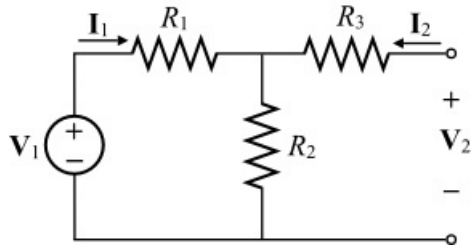


Figure 2

Step 4 of 6

Apply Kirchhoff's voltage law to left side loop.

$$V_1 = I_1 R_1 + I_1 R_2$$

$$I_1 = \frac{V_1}{R_1 + R_2} \dots\dots (5)$$

$$\frac{I_1}{V_1} = \frac{1}{R_1 + R_2}$$

$$g_{11} = \frac{1}{R_1 + R_2}$$

Step 5 of 6

Apply Kirchhoff's voltage law to left side loop.

$$V_2 = I_1 R_2$$

Substitute $\frac{V_1}{R_1 + R_2}$ for I_1 in $V_2 = I_1 R_2$.

$$V_2 = \left(\frac{V_1}{R_1 + R_2} \right) R_2$$

$$\frac{V_2}{V_1} = \frac{R_2}{R_1 + R_2}$$

$$g_{21} = \frac{R_2}{R_1 + R_2}$$

Step 6 of 6

To calculate parameters g_{12} and g_{22} , connect a test voltage source to the output port and short circuit the input port.

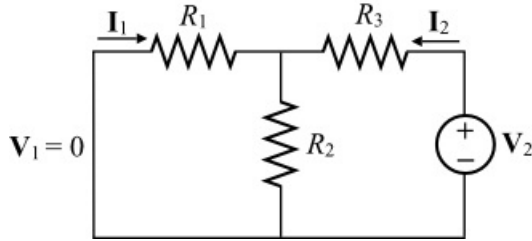


Figure 3

Apply Kirchhoff's voltage law to left side loop.

$$I_1 R_1 + (I_1 + I_2) R_2 = 0$$

$$I_1 = -\frac{I_2 R_2}{R_1 + R_2} \dots\dots (6)$$

$$\frac{I_1}{I_2} = -\frac{R_2}{R_1 + R_2}$$

$$g_{12} = -\frac{R_2}{R_1 + R_2}$$

Apply Kirchhoff's voltage law to right side loop.

$$\begin{aligned} V_2 &= I_2 R_3 + (I_2 + I_1) R_2 \\ &= I_1 R_2 + (R_2 + R_3) I_2 \end{aligned}$$

Substitute $-\frac{I_2 R_2}{R_1 + R_2}$ for I_1 in the above equation.

$$V_2 = \left(-\frac{I_2 R_2}{R_1 + R_2} \right) R_2 + (R_2 + R_3) I_2$$

$$\frac{V_2}{I_2} = -\frac{R_2^2}{R_1 + R_2} + R_2 + R_3$$

$$g_{22} = \frac{R_2^2 + R_1 R_2 - R_2^2}{R_1 + R_2} + R_3$$

$$g_{22} = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

Therefore, the g-parameters, $\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ are $\begin{bmatrix} \frac{1}{R_1 + R_2} & -\frac{R_2}{R_1 + R_2} \\ \frac{R_2}{R_1 + R_2} & \frac{R_1 R_2}{R_1 + R_2} + R_3 \end{bmatrix}$.